

EC442: Lecture Notes

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Notes on the 1st year Macro class

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1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'(c_t^i) = \alpha_j u'(c_t^j) \Rightarrow \frac{u'(c_t^i)}{u'(c_t^j)} = \frac{\alpha_j}{\alpha_i}. \quad (1.1)$$

Marginal-utility ratio constant across time/states.

- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j} \right)^{\frac{1}{\sigma}}. \quad (1.2)$$

Fixed consumption shares $\rightarrow$ **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights $\rightarrow$ unequal consumption.

- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity $\uparrow \rightarrow$ labor $\uparrow$; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

Lecture addition: Intuition for Complete Markets

- **Example:** 18 students; one high earner, others lower. Complete markets $\rightarrow$ high earner sells state-contingent claims $\rightarrow$ consumption decoupled from income.
- **Marx analogy:** "From each according to ability, to each according to needs." Efficient allocation equalizes marginal utilities regardless of income.
- **Empirical mismatch:** Data shows strong income-consumption correlation. Complete markets imply correlation vanishes $\rightarrow$ model unrealistic. Heterogeneity requires market incompleteness.

Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous:** assume some markets absent.
 1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous:** derive market incompleteness from frictions.
 1. e.g. limited commitment (participation constraints), private information.
 2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods:** Dynamic programming, numerical methods.
- **Background:** Equilibrium and welfare theorems.
- **Growth:** Basic models of economic growth.
- **Incomplete markets:** Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment:** Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information:** Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t. \quad (1.3)$$

- **Lagrangian;** current-value multipliers λ_t :

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1 - \delta)k_t - c_t - k_{t+1})]. \quad (1.4)$$

- **Interior FOCs:**

$$\begin{aligned} c_t : u'(c_t) &= \lambda_t, \\ k_{t+1} : \lambda_t &= \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \quad (1.5)$$

Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \quad (1.6)$$

- **Transversality condition:**

$$\lim_{T \rightarrow \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0. \quad (1.7)$$

Discounted shadow value of terminal capital $\rightarrow 0$.

- Increasing, concave u ; concave f ; finite discounted utility: feasibility + FOCs + TVC $\rightarrow$ global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0 :

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \quad (1.8)$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t \geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \quad (1.9)$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1 - \delta)k_0$.
- Inner max: given chosen k_1 , same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 $\rightarrow$ inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k , next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \quad (1.10)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.11)$$

Infinite sequence $\rightarrow$ one-period choice + discounted continuation value. Stationarity $\rightarrow$ same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:

- **Existence:** conditions for solution v ?
- **Uniqueness:** conditions for unique v within chosen function class?
- **Equivalence:** when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:

- *Recursive Macroeconomic Theory* (RMT), “Dynamic Programming” ([Ljungqvist & Sargent, 2018, ch. 3](#)).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Leads to 2nd-order difference eq. Solving via “shooting” → local, clunky.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: VFI guaranteed to converge to unique fixed point (via CMT), unlike many ODE/PDE methods.

Recording: 01:06:37, 01:07:39

1.5. Dynamic Programming

Bellman operator T : candidate continuation value $v \rightarrow$ updated value Tv .

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \quad (1.12)$$

subject to

$$c + k' = f(k) + (1 - \delta)k, c, k' \geq 0. \quad (1.13)$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \quad (1.14)$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \rightarrow S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \quad (1.15)$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \dots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \quad (1.16)$$

Hence $T^n v_0 \rightarrow v$ as $n \rightarrow \infty$, with geometric convergence. Here T^n denotes n successive applications of T , with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- Analogous to Brouwer fixed-point theorem in 1D: continuous function mapping interval to itself with slope $< 1 \rightarrow$ unique fixed point.
- In function spaces, “slope” = contraction factor β . Iterating operator moves points closer $\rightarrow$ convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K .
- $S = C(K)$: continuous real-valued functions on K ; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \quad (1.17)$$

(S, d) complete.

- Must verify $T : S \rightarrow S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \quad (1.18)$$

Theorem 1.6.1 $\rightarrow$ unique fixed point in S ; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- **Initial guess** $v^0 = 0$: One-period model (no future utility). Agent consumes all output today.
- **First iteration** Tv^0 : Two-period model. Optimize today/tomorrow, world ends after tomorrow.
- **Convergence**: Each iteration extends horizon by 1 period. $n \rightarrow \infty \rightarrow$ infinite-horizon value function.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u , f , and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \quad (1.19)$$

over feasible k' .

- **FOC for k'** :

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \quad (1.20)$$

- **Envelope condition**: differentiate w.r.t. k , holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \quad (1.21)$$

- **Combine**: apply envelope condition next period $\rightarrow$ usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \quad (1.22)$$

Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, \dots, k^N\} \subset K. \quad (1.23)$$

- Approximate value function by vector

$$\mathbf{v} = (v_1, v_2, \dots, v_N)^\top \in \mathbb{R}^N, v_i \approx v(k^i). \quad (1.24)$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \{k^1, \dots, k^N\}$. For each pair (i, j) , compute

$$c_{ij} = f(k^i) + (1 - \delta)k^i - k^j. \quad (1.25)$$

Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u(c_{ij})$. Each J_i must be nonempty.

2. **Initial guess:** any $\mathbf{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update:** for every $i = 1, \dots, N$,

$$v_i^{m+1} = \max_{j \in J_i} [u(c_{ij}) + \beta v_j^m]. \quad (1.26)$$

Use the old vector for all updates: $\mathbf{v}^{m+1} = T_N \mathbf{v}^m$.

4. **Convergence check:**

$$\Delta_m = \max_{1 \leq i \leq N} |v_i^{m+1} - v_i^m|. \quad (1.27)$$

If $\Delta_m \leq \varepsilon$, stop and return $\hat{\mathbf{v}} = \mathbf{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.

5. **Recover approximate policy:** for each i , choose

$$j_i \in \arg \max_{j \in J_i} [u(c_{ij}) + \beta \hat{v}_j]. \quad (1.28)$$

Set $k'(k^i) = k^{j_i}$ and $c(k^i) = c_{ij_i}$.

Why it converges: T_N is a β -contraction in the sup norm. **Theorem 1.6.1** $\rightarrow$ unique grid fixed point $\mathbf{v}_N^*$; **Algorithm 1.8.1** converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\mathbf{v}} - \mathbf{v}_N^*\|_\infty \leq \frac{\beta}{1 - \beta} \Delta_m \leq \frac{\beta}{1 - \beta} \varepsilon. \quad (1.29)$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- CMT guarantees convergence for individual decision problems (value functions).
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- **Frontier approaches**: Sparse grids; mean field games [ASR uncertain: Ben Moll]; heuristic trust in convergence when rigorous proofs unavailable.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not “fully forward-looking” as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

2. Lecture 2 Oct 5

2.1. Recall

2.1.1. Bellman equation for the standard growth model

$$v(k) = \max_{c,k'} [u(c) + \beta v(k')], u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad (2.1)$$

subject to

$$c + k' = (1 - \delta)k + k^\alpha, c > 0, k' \geq 0. \quad (2.2)$$

For $\sigma = 1$, use $u(c) = \log c$.

2.1.2. Contraction mapping theorem

On a nonempty complete metric space (S, d) , a contraction $T : S \rightarrow S$ with modulus $\beta \in (0, 1)$ has a unique fixed point v ([Theorem 1.6.1](#)):

$$\begin{aligned} d(Tx, Ty) &\leq \beta d(x, y), \\ d(T^n v_0, v) &\leq \beta^n d(v_0, v). \end{aligned} \quad (2.3)$$

Iteration from any $v_0 \in S$ converges geometrically to v .

2.2. Blackwell's sufficient conditions

Theorem 2.2.1 (Blackwell's sufficient conditions): Let $B(X)$ be the bounded real-valued functions on X , equipped with the sup norm, and let $T : B(X) \rightarrow B(X)$. Suppose:

1. **Monotonicity:** For all $f, g \in B(X)$,

$$f(x) \geq g(x) \text{ for all } x \in X \implies (Tf)(x) \geq (Tg)(x) \text{ for all } x \in X. \quad (2.4)$$

2. **Discounting:** There exists $\beta \in (0, 1)$ such that, for every $f \in B(X)$ and $a \geq 0$,

$$(T(f + a))(x) \leq (Tf)(x) + \beta a \text{ for all } x \in X. \quad (2.5)$$

Here $(f + a)(x) = f(x) + a$.

Then T is a contraction with modulus β :

$$\|Tf - Tg\|_\infty \leq \beta \|f - g\|_\infty. \quad (2.6)$$

2.2.1. Verification for the Bellman operator

Maximization preserves pointwise inequalities, so $f \geq g \implies Tf \geq Tg$. Adding a constant to continuation utility gives

$$(T(f + a))(k) = \max_{c,k'} [u(c) + \beta f(k')] + \beta a = (Tf)(k) + \beta a. \quad (2.7)$$

Both conditions hold; the self-map requirement $T : B(X) \rightarrow B(X)$ must also be verified.

2.3. Blackwell's conditions for the standard model

Let $\alpha, \delta, \beta \in (0, 1)$. Resources and consumption:

$$y(k) = \underbrace{(1 - \delta)k}_{\text{remaining capital}} + \underbrace{k^\alpha}_{\text{output}}, \quad (2.8)$$

$$c(k, k') = y(k) - k'.$$

The invariant state space satisfies

$$\bar{k} = \delta^{-\frac{1}{1-\alpha}},$$

$$X = [0, \bar{k}], \quad (2.9)$$

$$0 \leq y(k) \leq y(\bar{k}) = \bar{k} \text{ for all } k \in X.$$

Lecture addition: The capital ceiling

$\bar{k}$ is the steady state with zero consumption: all output goes into investment, and $\bar{k}^\alpha = \delta\bar{k}$. The bound comes from feasibility alone. Starting in X , feasible savings remain in X .

Recording: 00:07:02 - 00:07:39

Feasible savings and the Bellman operator:

$$\Gamma(k) = \{k' \geq 0 : c(k, k') \geq 0\} \subseteq X,$$

$$(Tf)(k) = \sup_{k' \in \Gamma(k)} \left[\underbrace{u(c(k, k'))}_{\text{current utility}} + \beta \underbrace{f(k')}_{\text{continuation value}} \right]. \quad (2.10)$$

Extend u to $c = 0$ by its limit, possibly $-\infty$. Replace $\sup$ with $\max$ when the optimum is attained.

2.3.1. Monotonicity

For $f, g \in B(X)$ with $f \geq g$ pointwise,

$$\begin{aligned} (Tf)(k) &= \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta f(k')] \\ &\geq \sup_{k' \in \Gamma(k)} [u(c(k, k')) + \beta g(k')] \\ &= (Tg)(k). \end{aligned} \quad (2.11)$$

Lecture addition: Why the first inequality can be strict

If the maximum for g is attained, let $k^*(k)$ be its optimal savings:

$$\begin{aligned}(Tf)(k) &\geq u(c(k, k^*(k))) + \beta f(k^*(k)), \\ &\geq u(c(k, k^*(k))) + \beta g(k^*(k)), \\ &= (Tg)(k).\end{aligned}$$

The first step evaluates one feasible choice. It can be strict because $k^*(k)$ need not maximize the objective with continuation value f .

Recording: 00:08:00 - 00:10:59

2.3.2. Discounting

For $f \in B(X)$ and a constant $a \geq 0$,

$$\begin{aligned}(T(f+a))(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k, k')) + \beta f(k') + \underbrace{\beta a}_{\text{constant in } k'} \right] \\ &= (Tf)(k) + \beta a.\end{aligned}\tag{2.12}$$

2.3.3. Boundedness and conclusion

For $0 < \sigma < 1$, $u(0) = 0$ and

$$\begin{aligned}0 \leq u(c(k, k')) &\leq u(\bar{k}) < \infty, \\ \|Tf\|_\infty &\leq u(\bar{k}) + \beta \|f\|_\infty < \infty.\end{aligned}\tag{2.13}$$

Thus $T : B(X) \rightarrow B(X)$ on the complete sup-norm space $B(X)$. By [Theorem 2.2.1](#) and [Theorem 1.6.1](#), there is a unique $v \in B(X)$ with $Tv = v$, and for every $v_0 \in B(X)$,

$$\|T^n v_0 - v\|_\infty \leq \beta^n \|v_0 - v\|_\infty \rightarrow 0.\tag{2.14}$$

For $\sigma \geq 1$ (including log utility),

$$k = 0 \implies \Gamma(k) = \{0\} \implies (Tf)(0) = -\infty.\tag{2.15}$$

The bounded-function theorem then requires a different state or function space.

2.4. Formulating recursive models

Identify the state variables before writing the Bellman equation.

A **state variable**:

- Matters for the decision problem.
- Is taken as given by the decision maker at the time of choice.
- Varies over time or across states.

Other variables:

- **Choice variables**: chosen by the decision maker.

- **Parameters:** affect the problem but are fixed over time and across states.

The value function takes the state variables as its arguments.

Lecture addition: A past choice can be a current state

The household chose today's capital in the past; it is fixed when today's consumption-saving decision is made. Models with habits or varying patience may require preference states. Here β and σ remain parameters because they are fixed.

Recording: 00:16:22 - 00:18:54

2.4.1. How to determine the state

1. **Fix timing:** Specify what is observed before choices are made. A lagged choice may be today's state; a newly chosen quantity is a control.
2. **List candidates:** Retain inherited stocks and observed shocks that affect current constraints, utility, or forecasts. Separate fixed parameters.
3. **Check sufficiency:** Given state s and choices q , specify

$$\begin{aligned} q &\in \Gamma(s), \\ \text{current utility} &= U(s, q), \\ s' &= G(s, q, \varepsilon'). \end{aligned} \tag{2.16}$$

Here ε' is future uncertainty. Two histories with the same s must give the same feasible choices, current utility, and conditional distribution of s' for every feasible q .

4. **Remove redundancy:** Drop a coordinate only if this sufficiency test still holds. A one-to-one change of coordinates gives an equivalent state.
5. **Write the Bellman equation:**

$$v(s) = \sup_{q \in \Gamma(s)} \left[\underbrace{U(s, q)}_{\text{current utility}} + \beta \underbrace{\mathbb{E}[v(s') | s, q]}_{\text{expected continuation}} \right]. \tag{2.17}$$

Use max when the optimum is attained; omit the expectation in a deterministic model.

Reduction test: If two candidate states have the same proposed summary but different feasible choices, utility, or transition distributions, the summary is insufficient.

2.4.2. Example: standard model with two sectors

Each period, allocate inherited capital k and one unit of labor freely between consumption (c) and investment (i) sectors. Let $\alpha, \gamma \in (0, 1)$.

Feasibility:

$$\begin{aligned}
 c &= \underbrace{k_c^\alpha n_c^{1-\alpha}}_{\text{consumption output}}, \\
 k' &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
 k_c + k_i &= k, \\
 n_c + n_i &= 1.
 \end{aligned} \tag{2.18}$$

All quantities are nonnegative.

State identification:

- k is inherited; (k_c, k_i, n_c, n_i) are chosen before production.
- Free reallocation removes dependence on the previous sectoral split.
- Given k and current choices, utility and next-period k' are determined.

Hence $s = k$ is sufficient.

The Bellman equation is

$$v(k) = \max_{c, k', k_c, k_i, n_c, n_i} [u(c) + \beta v(k')], \tag{2.19}$$

subject to the constraints above, with u as defined in Recall.

Lecture addition: Timing and the sectoral split

Let $\tilde{v}(k_c, k_i)$ be the value after capital has been allocated across sectors. Before allocation,

$$v(k) = \max_{k_c + k_i = k} \tilde{v}(k_c, k_i).$$

In this deterministic model, the future split can be planned a period earlier. Treating the initial allocation consistently gives the same choices. With M grid points per state, retaining both stocks requires M^2 combinations; retaining total capital requires M .

Recording: 00:24:30 - 00:30:44

2.4.3. Example: two sectors with irreversible investment

Capital is installed before the period and cannot move between sectors. Choose labor allocations and new investment x_c, x_i today; investment becomes productive next period.

Feasibility:

$$\begin{aligned}
 c &= k_c^\alpha n_c^{1-\alpha}, \\
 x_c + x_i &= \underbrace{k_i^\gamma n_i^{1-\gamma}}_{\text{investment output}}, \\
 k'_c &= (1 - \delta)k_c + x_c, \\
 k'_i &= (1 - \delta)k_i + x_i, \\
 n_c + n_i &= 1, \\
 x_c, x_i &\geq 0.
 \end{aligned} \tag{2.20}$$

All other quantities are nonnegative. Irreversibility implies

$$k'_j \geq (1 - \delta)k_j \text{ for } j \in \{c, i\}. \tag{2.21}$$

Sectoral capital can decline only through depreciation.

Lecture addition: Why the aggregate equation is insufficient

The state is (k_c, k_i) . Adding the accumulation equations leaves the separate restrictions $x_c, x_i \geq 0$ in place. Dropping them would permit conversion of existing capital between sectors. They can bind when initial stocks are imbalanced or shocks change the desired sectoral mix.

Recording: 00:32:20 - 00:34:45

Lecture addition: Applications of multisector models

Separate consumption and investment sectors allow general and investment-specific technological change to have different effects on business cycles. Larger models use input-output networks, with sectors buying CES aggregates of intermediate inputs, to study shock propagation.

Recording: 00:20:42 - 00:20:53; 00:35:00 - 00:36:15

2.4.4. Standard model with shocks

Productivity z is i.i.d. over time. The resource constraint is

$$c + k' = (1 - \delta)k + zk^\alpha. \tag{2.22}$$

In all formulations, $c, k' \geq 0$.

2.4.4.1. States (k, z) : after the shock

Observe z , then choose c and k' :

$$v(k, z) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [v(k', z')]], \tag{2.23}$$

subject to [Equation \(2.22\)](#). The expectation is over next period's shock z' .

2.4.4.2. State x : total resources

After observing z , define

$$x = (1 - \delta)k + zk^\alpha. \quad (2.24)$$

Then

$$w(x) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'} [w(x')]], \quad (2.25)$$

subject to

$$\begin{aligned} c + k' &= x, \\ x' &= (1 - \delta)k' + z'(k')^\alpha. \end{aligned} \quad (2.26)$$

z' is independent of z , so the transition of x' requires only k' .

Note: This reduction to x relies on z being i.i.d. over time.

Lecture addition: The forecast matters with persistent shocks

Two states can have the same current resources x but different productivity z . With persistence, their distributions of z' differ, so their optimal savings can differ. Current z carries information about tomorrow.

Recording: 00:45:42 - 00:46:22

2.4.4.3. State k : before the shock

$V(k)$ is the value *before* z is observed. Choices $c(z)$ and $k'(z)$ may depend on its realization:

Choices. For a fixed (k, z) or x , Options 1 and 2 maximize over scalar choices c, k' . Their policy functions are $c(k, z), k'(k, z)$ or $c(x), k'(x)$; state arguments are implicit. Option 3 maximizes over functions $c(z), k'(z)$ for every possible z .

$$V(k) = \max_{c(z), k'(z)} \mathbb{E}_z [u(c(z)) + \beta V(k'(z))], \quad (2.27)$$

subject to

$$c(z) + k'(z) = (1 - \delta)k + zk^\alpha \text{ for every } z. \quad (2.28)$$

$$\begin{aligned} v(k, z) &= w((1 - \delta)k + zk^\alpha), \\ V(k) &= \mathbb{E}_z [v(k, z)]. \end{aligned} \quad (2.29)$$

Timing. Before observing z , $V(k)$ averages over its realization; i.i.d. shocks make past productivity uninformative about z . After observing z , the state records realized resources through (k, z) or x . Decisions can depend on observed z in all three formulations.

Lecture addition: Several values within one period

A model can use value functions at several stages of a period. A value before information arrives is linked to the later value by an expectation over the information revealed between those stages.

Recording: 00:51:25 - 00:52:16

2.5. Equilibrium

A social planner maximizes welfare subject to feasibility constraints. In a decentralized economy, agents make their own choices, linked through markets, institutions, and policies.

- How can decentralized outcomes be represented formally?
- How do equilibrium allocations relate to the planner's solution?
- Can decentralized equilibria be formulated recursively?

2.6. What is a competitive equilibrium?

A competitive equilibrium consists of:

- An *allocation*: consumption, production, and factor inputs.
- A *price system*: prices in every market.

Two conditions:

- **Optimization**: Households maximize utility and firms maximize profits, taking prices as given.
- **Feasibility**: The allocation is feasible: total use of each good does not exceed production plus endowments.

Lecture addition: Trading dates and events

With complete markets, goods are distinguished by delivery date and event. The lecture's example is consumption in 2027 conditional on no rain in Australia. Suitable sequential asset markets can span the same contingent consumption trades as markets opened once at date 0.

Recording: 01:02:26 - 01:03:56

2.7. Sequential equilibrium

In the deterministic standard model, markets open each period. The household owns capital k_t and a fixed labor endowment $n_t = 1$. It rents capital and supplies labor to the firm; both agents take prices as given. Let r_t be the net return on capital. The capital rental rate is $r_t + \delta$; the wage is w_t .

Given k_0 , a sequential equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ satisfying the following conditions.

2.7.1. Household

Given the price paths, the household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.30)$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= \underbrace{(1 + r_t)k_t}_{\text{capital payoff}} + \underbrace{w_t n_t}_{\text{labor income}}, \\ n_t &= 1, \\ c_t &\geq 0, \\ k_{t+1} &\geq 0. \end{aligned} \quad (2.31)$$

2.7.2. Firm

The firm chooses capital demand K_t and labor demand N_t each period:

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.32)$$

Lecture addition: Representative agents and price taking

The representative firm behaves as an atomistic price taker. Its labor demand N_t is a choice; $N_t = 1$ is imposed through market clearing. Representative notation can stand for a continuum of identical agents. Separate demand and supply symbols make this distinction explicit.

Recording: 01:06:36 - 01:08:06; 01:12:24 - 01:13:53

2.7.3. Market clearing

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.33)$$

At positive inputs, the firm's first-order conditions imply

$$\begin{aligned} r_t &= \alpha k_t^{\alpha-1} - \delta, \\ w_t &= (1 - \alpha)k_t^\alpha. \end{aligned} \quad (2.34)$$

Lecture addition: Depreciation and the quoted return

An alternative convention quotes the capital rental rate $\rho_t = r_t + \delta$ and lets the household bear depreciation. Then

$$\begin{aligned} c_t + k_{t+1} &= (1 - \delta + \rho_t)k_t + w_t n_t, \\ \text{firm capital cost} &= \rho_t K_t. \end{aligned}$$

The net-return and rental-rate conventions describe the same payments.

Recording: 01:10:41 - 01:11:28; 01:14:52 - 01:16:21

2.7.4. No-Ponzi condition

If borrowing is allowed, replace $k_{t+1} \geq 0$ with

$$\liminf_{t \rightarrow \infty} \frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} \geq 0. \quad (2.35)$$

At an interior optimum with $u'(c_s) > 0$, the Euler equation gives

$$1 + r_s = \frac{u'(c_{s-1})}{\beta u'(c_s)}. \quad (2.36)$$

Multiplying from $s = 1$ to t , marginal utilities cancel:

$$\begin{aligned} \prod_{s=1}^t (1 + r_s) &= \prod_{s=1}^t \frac{u'(c_{s-1})}{\beta u'(c_s)}, \\ &= \frac{u'(c_0)}{\beta^t u'(c_t)}. \end{aligned} \quad (2.37)$$

Hence

$$\frac{k_{t+1}}{\prod_{s=1}^t (1 + r_s)} = \frac{\beta^t u'(c_t) k_{t+1}}{u'(c_0)}. \quad (2.38)$$

Since $u'(c_0) > 0$ is constant, dividing by it preserves the sign of the limit inferior. Thus the no-Ponzi condition is equivalent to

$$\liminf_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} \geq 0. \quad (2.39)$$

The household cannot finance consumption by rolling over debt forever.

Lecture addition: Physical capital and financial assets

Physical capital is nonnegative. An individual's financial asset position can be negative when borrowing is allowed for consumption smoothing. Period budgets alone permit arbitrarily large debt followed by refinancing of principal and interest. A borrowing limit or no-Ponzi condition restricts those plans.

Recording: 01:16:28 - 01:17:20; 01:18:29 - 01:19:47

2.8. Arrow-Debreu equilibrium

Sequential markets open each period. In an Arrow-Debreu equilibrium, markets open once at date 0. Agents contract for goods delivered at each future date (and state, under uncertainty). Delivery and production still occur each period.

Given k_0 , an Arrow-Debreu equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, r_t, w_t\}_{t=0}^{\infty}$ satisfying the conditions below.

The additional price $p_t > 0$ is the date-0 price of one unit of the good delivered at date t , with $p_0 = 1$. It converts date- t quantities into date-0 goods. The wage w_t and capital rental rate $r_t + \delta$ remain measured in date- t goods. Their date-0 prices are $p_t w_t$ and $p_t(r_t + \delta)$.

2.8.1. Household

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2.40)$$

subject to $n_t = 1$, $c_t \geq 0$, given k_0 , and a single present-value budget:

$$\sum_{t=0}^{\infty} p_t(c_t + k_{t+1}) = \sum_{t=0}^{\infty} p_t[(1 + r_t)k_t + w_t n_t]. \quad (2.41)$$

The present-value sums are assumed finite.

2.8.2. Firm and market clearing

For each delivery date t , the firm solves

$$\max_{K_t, N_t \geq 0} [K_t^\alpha N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t]. \quad (2.42)$$

Multiplying profits by $p_t > 0$ leaves the input choices unchanged. Markets clear at every date:

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^\alpha. \end{aligned} \quad (2.43)$$

2.8.3. No arbitrage and the present-value budget

In the household's budget, one unit of k_{t+1} costs p_t and generates a payoff worth $p_{t+1}(1 + r_{t+1})$ at date 0. No arbitrage requires

$$\begin{aligned} p_t - p_{t+1}(1 + r_{t+1}) &= 0, \\ \frac{p_t}{p_{t+1}} &= 1 + r_{t+1}. \end{aligned} \quad (2.44)$$

Thus $\frac{p_t}{p_{t+1}}$ is the gross real return from t to $t + 1$. With $p_0 = 1$,

$$p_t = \frac{1}{\prod_{s=1}^t (1 + r_s)}. \quad (2.45)$$

Capital purchases and their future payoffs cancel from the lifetime budget:

$$\sum_{t=0}^{\infty} p_t c_t = \underbrace{(1 + r_0)k_0}_{\text{initial capital wealth}} + \underbrace{\sum_{t=0}^{\infty} p_t w_t}_{\text{lifetime labor income}}. \quad (2.46)$$

The lifetime budget makes a separate no-Ponzi condition unnecessary.

Bibliography

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